

ALI NIKKHAH

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SUMMARY

ML Platform Engineer with 2+ years building scalable ML infrastructure: Kubernetes-native platforms, LLM serving stacks (vLLM, Triton), distributed training (Ray), MLOps pipelines (MLflow, ArgoCD), and high-concurrency model serving (FastAPI, TensorFlow Serving). At Digikala, engineered the LLM serving layer for 5M+ daily queries: vLLM PagedAttention, Triton, multi-tier Redis, async FastAPI, K8s multi-tenant (200 pods). At HomaCloud, founded ML platform from scratch: Ray + MLflow + TF Serving, CI/CD ;10min, 15+ models. At Turquoise, built data lake (Trino, PeerDB CDC ;60s) and ML model deployment. Expert in: ML infrastructure, GPU scheduling, model optimization, platform engineering.

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

ML Platform Engineer (LLM Serving)

Jan 2025 – Aug 2025

- Engineered LLM serving stack for 5M+ daily queries: vLLM PagedAttention + Triton Inference Server, async FastAPI, multi-tier Redis caching, request batching achieving **3.2x throughput, 60% cost reduction, p99 ;200ms**.
- Designed Modular Control Pipeline (MCP) for zero-downtime model hot-swapping and live index migrations; integrated LangGraph for multi-agent session management.
- Architected multi-tenant K8s deployment (200 pods) with GPU scheduling (NVIDIA device plugin, MIG) isolating vision/language clusters (**40% GPU cost reduction**); implemented HPA/VPA for burst traffic.

Advanced Analytics Australia

Remote Contract

ML Platform Engineer (Edge)

Sep 2025 – Apr 2026

- Built edge ML serving infrastructure: TensorRT/TF Lite compilation, Triton Inference Server, Kafka streaming for **100+ cameras @ 30 FPS**; dynamic batching eliminating Python overhead (**3.5x speedup**).
- Implemented edge MLOps: zero-downtime model updates across 12 distributed sites; Prometheus/Grafana monitoring for inference health, drift detection, hardware failures.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer

Sep 2025 – Mar 2026

- Architected enterprise data lake: Trino + PeerDB CDC (**;60s latency, 5TB/day**); Airflow DAGs for multi-stage transformations; PySpark ETL (**60% memory reduction**); Iceberg/Parquet on object storage.
- Deployed ML models for credit risk (AUC 0.87) and portfolio allocation; built FastAPI inference APIs with model versioning, A/B testing (mSPRT), and automated rollback.

HomaCloud

Tehran, Iran

Founding MLOps Engineer

Aug 2021 – Feb 2022

- Founded ML platform from scratch: K8s + Ray (distributed training) + MLflow (experiment tracking) + TensorFlow Serving; CI/CD (ArgoCD, GitHub Actions, Terraform **;10min releases**).
- Established ML best practices: model registry, lineage tracking, automated testing, canary deployments; **15+ models, 99.9% deploy success**; trained team on MLOps.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

RESEARCH HIGHLIGHTS

Univ. of British Columbia (*Remote RA, Feb 2024 – Nov 2025*) — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center (*Remote RA, Apr 2023 – Feb 2024*) — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.